Unit 8: Hypothesis Testing for one population.

Topics: Hypothesis Testing for Population Mean (μ) and Population Proportion (p)

1. Introduction to Hypothesis Testing

Hypothesis testing is a formal procedure used to decide whether sample data provides sufficient evidence to reject a claim about a population parameter. In computing, we use hypothesis tests constantly: to determine whether a new algorithm is faster, whether a server's response time has changed, or whether a software update improved user retention.

1.1 Core Concepts

Key Definitions

Null Hypothesis (H_0): The default assumption about the population parameter. It represents 'no change' or 'no effect'. We assume it is true until evidence suggests otherwise.
Alternative Hypothesis (H_1 or H_a): What we are trying to find evidence for. It contradicts H_0.
Test Statistic: A value calculated from sample data that is used to assess how far the sample result is from H_0.
p-value: The probability of observing a test statistic same as (or more extreme than) the one computed, assuming H_0 is true.
Significance Level (α): The threshold for the p-value. Common choices: $\alpha = 0.05$ or $\alpha = 0.01$.

1.2 Decision Rule (p-value Method)

This course uses the p-value method exclusively:

Decision Rule

If p-value ≤ α ⇒ Reject H_0 (sufficient evidence to support H)
If p-value > α ⇒ Fail to Reject H_0 (insufficient evidence to support H)

Important: 'Fail to reject H_0' does NOT mean H_0 is true. It simply means we lack enough evidence to reject it.

1.3 Types of Tests (by Tail)

Test Type	Alternative Hypothesis & Use Case
Two-Tailed	H_A : $\mu \neq \mu_0$ or $p \neq p_0$ | Use when testing if a parameter has changed in either direction.
Right-Tailed (Upper)	H_A: $\mu > \mu_0$ or $p > p_0$ | Use when testing if a parameter has increased.
Left-Tailed (Lower)	H_A: $\mu < \mu_0$ or $p < p_0$ | Use when testing if a parameter has decreased.

1.4 The 5-Step Hypothesis Testing Procedure

We follow this structured approach for every test in this course:

1. State the hypotheses: Write H_0 and H using proper notation.
2. Identify the significance level (α).
3. Compute the test statistic and p-value using RStudio.
4. Make a decision: Compare p-value to α.
5. State a conclusion in context: Interpret the result in plain language relevant to the problem.

2. Hypothesis Test for Population Mean (μ)

2.1 When to Use

- The population is normally distributed, OR the sample size is large (n ≥ 30).
- The population standard deviation (σ) is unknown → use a t-test with n − 1 degrees of freedom.
- The population standard deviation (σ) is known → use a z-test (rare in practice).

2.2 Test Statistic Formula

t-Test Statistic (most common — σ unknown)

$$t = \frac{(\bar{x} - \mu)}{\left(\frac{s}{\sqrt{n}}\right)}$$

Where: $\bar{x}$ = sample mean μ = hypothesized population mean
s = sample standard deviation n = sample size
Degrees of freedom: df = n − 1

2.3 RStudio Functions

The primary function for t-tests in R is t.test(). Key arguments:

Argument	Description
x	A numeric vector of sample data
mu	The hypothesized population mean (μ_0)
alternative	"two.sided" (default), "greater", or "less"
conf.level	Confidence level (default = 0.95, so α = 0.05)

2.4 Example 1: Two-Tailed Test for Mean

$\mu = 8.5$ s.

Scenario

A cloud provider advertises that its virtual machines boot in a mean of 8.5 seconds. A system administrator suspects the actual mean has changed (either faster or slower) after a recent infrastructure update. She collects a random sample of 35 boot times (in seconds) and records the following data. Use α = 0.05.

$n = 35$

RStudio Code and Output

```
# Cloud VM boot time data (seconds)
boot_times <- c(8.9, 7.8, 9.1, 8.3, 8.7, 9.4, 8.0, 8.6, 7.9, 9.2,
                8.5, 9.0, 8.4, 8.8, 7.6, 9.3, 8.1, 8.7, 9.5, 8.2,
                8.9, 7.7, 9.1, 8.6, 8.3, 9.0, 8.4, 8.8, 9.2, 8.5,
                8.7, 7.8, 9.3, 8.6, 8.4)

t.test(boot_times, mu = 8.5, alternative = "two.sided", conf.level = 0.95)
```

z_{score} of $\bar{x}$:

$$\frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}$$

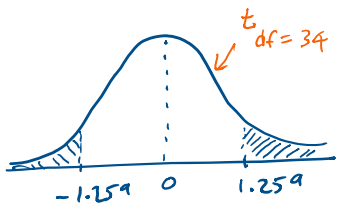
H_0: $\mu = 8.5$ s

H_A: $\mu \neq 8.5$ This is a two-tailed test

$\bar{x} = 8.609$

$n = 35$

$S = 0.512$



$$\text{test statistic} = t = \frac{(8.609 - \overset{\mu}{8.5})}{\left(\frac{0.512}{\sqrt{35}}\right)} = 1.259$$

p-value:

```
2*pt(-1.259,34)
[1] 0.216606
```

Decision: the p-value (22%) is not less than the 5% alpha (we are observing a common occurrence, not an unusual occurrence). We don't have enough reason to reject the cloud provider's claim. We fail to reject H0

Conclusion: At a 5% significance level, the true population mean boot time does not seem to have changed

2.5 Example 2: Right-Tailed (One-Tailed) Test for Mean

Scenario

A software development team claims their new build pipeline completes in a mean of 12 minutes. A DevOps engineer suspects that after adding new security scanning steps, builds now take longer than 12 minutes on average. He samples 28 recent build durations. Use α = 0.01.

$n = 28$

$\alpha = 1\%$

RStudio Code and Output

```
# Build pipeline completion times (minutes)
build_times <- c(13.2, 12.1, 14.5, 11.8, 13.7, 12.9, 14.1, 13.5, 12.6,
13.8,14.3, 12.4, 13.0, 12.7, 14.6, 13.1, 12.5, 13.9, 14.0, 12.8,
13.3, 12.2, 14.4, 13.6, 12.3, 13.4, 14.2, 13.0)

mean(build_times) = 13.28214
sd(build_times) = 0.7939837

# Right-tailed t-test: H0: mu = 12, H1: mu > 12
t.test(build_times, mu = 12, alternative = "greater", conf.level = 0.99)
```

$\bar{x} = 13.28214$ ← observed

$S = 0.79398$

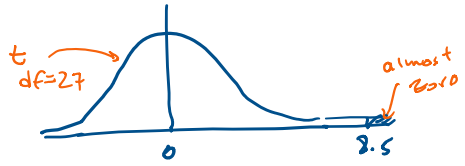
$H_0: \mu = 12$ minutes

$H_A: \mu > 12$ minutes

This is a right-tailed test

test statistic: $t = \dfrac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}} = \dfrac{(13.28214 - 12)}{\left(\frac{0.79398}{\sqrt{28}}\right)} = 8.5449$

p-value $= Pr(t \geq 8.5449) = 1 - pt(8.5449, 27)$ or $Pt(-8.5449, 27)$



p-value = 1.848269e-09 $\simeq \phi$

Decision: The p-value is almost zero, it is much less the 1% alpha. We have strong support from this sample to reject H0.

Conclusion: At a 1% significance level, the findings support that the addition of the new security scanning steps has increased the mean build time.

2.6 Example 3: Left-Tailed (One-Tailed) Test for Mean

Scenario

A company's old web server had a mean API response time of 250 milliseconds. After migrating to a new server, an engineer wants to confirm that response times have decreased (improved). A sample of 40 API calls is recorded. Use α = 0.05.

RStudio Code and Output

$\alpha = 5\%$ $n = 40$

```
# API response times in milliseconds
response_ms <- c(241, 238, 255, 243, 247, 236, 252, 244, 239, 248,
                 235, 251, 242, 246, 237, 253, 245, 240, 249, 234,
                 254, 243, 238, 247, 241, 250, 236, 244, 239, 248,
                 237, 252, 243, 246, 235, 251, 242, 238, 247, 240)

mean(response_ms) = 243.65
sd(response_ms) = 5.90328


# Left-tailed t-test: H0: mu = 250, H1: mu < 250
t.test(response_ms, mu = 250, alternative = "less", conf.level = 0.95)
```

$H_0: \mu = 250$ ms

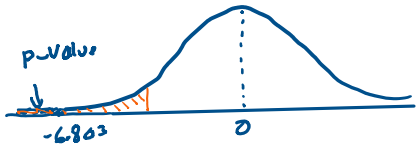
$H_A: \mu < 250$ ms

This is a left-tailed test

test statistic: $\dfrac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \dfrac{243.65 - 250}{\left(\frac{5.90328}{\sqrt{40}}\right)} = -6.803$

p-value $= P_r(t \leq -6.803)$

$= pt(-6.803, 39) =$ 1.998681e-08 $\simeq \phi$



Decision: the p-value is almost zero, much less than the 5% significance level. We have strong support from sample data to reject H0

Conclusion: At a 5% significance level the sample data confirms that the mean response time is decreased after the migration. :)

3. Hypothesis Test for Population Proportion (p)

3.1 When to Use

- The variable of interest is categorical (binary: success/failure, yes/no, pass/fail).
- Both np ≥ 10 and n(1 − p) ≥ 10 must hold (normal approximation condition).
- Use a z-test for proportions.

3.2 Test Statistic Formula

z-Test Statistic for Proportion

$$z = \frac{(\hat{p} - p)}{\sqrt{\frac{pq}{n}}} = \text{Z-score}$$

Where: $\hat{p}$ = sample proportion = x / n
p= hypothesized population proportion
x = number of successes n = sample size

3.3 RStudio Function

Use prop.test() in R.

3.4 Example 4: Two-Tailed Test for Proportion

Scenario

A software company historically has a 15% defect rate in code deployments. After introducing a new automated testing framework, the QA manager wants to know if the defect rate has changed (could be higher or lower). In a sample of 200 recent deployments, 22 had defects. Use α = 0.05.

$n = 200$

$\hat{p} = \frac{22}{200} = 11\% = 0.11$

$\begin{cases} H_0: p = 0.15 \\ H_A: p \neq 0.15 \end{cases}$

This is a two-tailed test

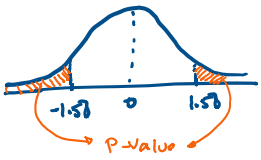
$\alpha = 5\%$

Normality Check

$200(0.15) = 30 > 10$ ✓

$200(0.85) = 170 > 10$ ✓

test statistic: $\frac{\hat{p} - p}{\sqrt{\left(\frac{pq}{n}\right)}} = \frac{(0.11 - 0.15)}{\sqrt{\left(\frac{0.15 \times 0.85}{200}\right)}} = -1.58 = z$



P-value = $Pr(z \leq -1.58 \text{ or } z \geq 1.58) = 2 * \text{pnorm}(-1.58)$
$= 0.113$

Decision: the p-value (11.3%) is not less than alpha = 5%, therefore nothing unusual has happened and we don't have enough reason to reject H0. We fail to reject H0.
Conclusion: At a 5% significance level, the defect rate does not seem to have changed.

μ $\quad$ p

t $\quad$ z

3.5 Example 5: Right-Tailed Test for Proportion

Scenario

Industry data shows that 60% of cybersecurity incidents ~~go undetected~~ get detected by traditional signature-based systems. A security team implements a new AI-based intrusion detection system and claims it detects more than 60% of incidents. In a controlled test, 78 out of 120 simulated attacks were detected. Use α = 0.05.

Normality check

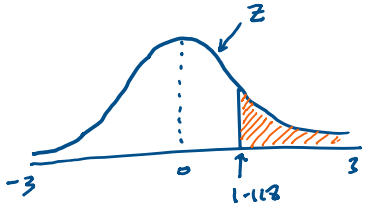
$(120)(0.60) \overset{?}{>} 10$ Yes

$(120)(0.40) \overset{?}{>} 10$ Yes

$H_0: p = 0.60$ $\quad$ This is a right-tailed test

$H_A: p > 0.60$ $\quad$ $\hat{p} = \frac{78}{120} = 0.65$

test statistic $z = \dfrac{\hat{p} - p}{\sqrt{\left(\frac{pq}{n}\right)}} = \dfrac{(0.65 - 0.60)}{\sqrt{\left(\frac{0.60 \times 0.40}{120}\right)}} = 1.118$



p-value : $Pr(z \geq 1.118) = 1 - \text{pnorm}(1.118)$

$= 0.1317835$

Decision: the p-value (13.2%) is much higher than the 5% alpha, therefore nothing unusual has happened and we don't have enough reasons to reject H0. We fail to reject H0

Conclusion: Based on evidence provided by this sample, at a 5% significance level, the AI-based system does not have a true proportion rate higher than 60% (it is not an actual improvement).

3.6 Example 6: Left-Tailed Test for Proportion

Scenario

A tech company previously had a 30% user churn rate per quarter. After launching a new onboarding experience, the product team hopes the churn rate has decreased. Out of 500 users who completed the new onboarding, 135 churned within the quarter. Use α = 0.01.

$H_0: p = 0.30$

$H_A: p < 0.30$

This is a left-tailed test

$n = 500$

$\hat{p} = \frac{135}{500} = 0.27$

$\alpha = 1\%$

Normality Check:

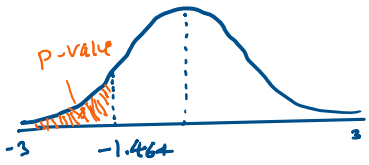
$(500)(0.30) = 150 \overset{?}{>} 10$ Yes

$(500)(0.70) = 350 \overset{?}{>} 10$ Yes

test statistic: $Z = \frac{\hat{p} - p}{\sqrt{\left(\frac{pq}{n}\right)}} = \frac{0.27 - 0.30}{\sqrt{\left(\frac{0.30 \times 0.70}{500}\right)}} = -1.464$

P-value $= Pr(Z \leq -1.464) = \text{Pnorm}(-1.464)$

$= 0.07159697$



Decision: The p-value (7.2%) is much greater than the 1% alpha. Nothing unusual has occurred. Therefore we fail to reject H0

Conclusion: At a 1% significance level we cannot say that the churn rate has decreased. Keep hoping!